

GRADUATION PROJECT · NETWORK & CYBERSECURITY

Enterprise Banking Network Security Architecture

Design and Implementation of a High-Availability, Multi-Site Banking Network with Centralized LDAP Authentication, Site-to-Site VPN, and SD-WAN

High Availability (HA)

LDAP Authentication

IPsec VPN

SD-WAN

FortiGate NGFW

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1. Abstract

This project presents the design and practical implementation of a secure, highly available, multi-site network architecture for a simulated enterprise banking environment. The network connects a Headquarters site in Egypt with two remote branches in Libya and Tunisia over a public Internet Service Provider (ISP) cloud, using FortiGate next-generation firewalls as the core security and routing platform.

The solution addresses four core enterprise networking and security requirements. First, **High Availability (HA)** is implemented at the Headquarters site using two FortiGate firewalls configured in an Active-Passive cluster, ensuring that a single device failure does not interrupt banking operations. Second, **centralized authentication** is achieved by integrating the FortiGate firewalls with a Lightweight Directory Access Protocol (LDAP) server hosted on the internal Active Directory (AD) domain controller, allowing user identity and group-based access policies to be managed from a single source of truth. Third, **site-to-site connectivity** between the three locations is secured using a Hub-and-Spoke IPsec VPN topology, encrypting all inter-branch banking traffic as it traverses the public ISP cloud. Finally, **SD-WAN** is configured on the branch firewalls to continuously monitor link performance (latency, jitter, and packet loss) and steer traffic intelligently, improving resilience against WAN link degradation.

The entire architecture was built and validated in a GNS3 virtual lab using FortiGate VM64-KVM firewall appliances, VPCS virtual clients, and SecureCRT for console/terminal access. Functional testing — including cross-site ping tests, VPN tunnel status verification, HA failover monitoring, SD-WAN performance SLA checks, and LDAP-based user login — confirms that the network meets its design goals of availability, confidentiality, and centralized control, all of which are critical requirements for a financial institution's IT infrastructure.

2. Introduction

2.1 Project Background

Banks and financial institutions operate distributed networks that must remain online at all times, protect sensitive customer and transaction data, and give IT administrators strict, centralized control over who can access what. A single point of failure, an unencrypted branch link, or unmanaged local user accounts can all translate into direct financial and reputational risk. This project simulates a realistic banking network spanning a Headquarters site and two international branches, and applies enterprise-grade security controls that a real bank's network security team would be expected to design and operate.

2.2 Objectives

- Design a multi-site banking network topology connecting a Headquarters site (Egypt) with two remote branches (Libya and Tunisia) through a shared ISP cloud.
- Eliminate the single point of failure at Headquarters by deploying a two-node FortiGate **High Availability** cluster in Active-Passive mode.
- Centralize user authentication and access control by integrating FortiGate with an **LDAP** / Active Directory server, and enforce group-based firewall policies.
- Secure all inter-site banking traffic using an encrypted **Hub-and-Spoke IPsec VPN**.
- Improve WAN resilience and visibility using **SD-WAN** performance monitoring and traffic steering on the branch firewalls.
- Validate the design through practical connectivity, failover, authentication, and performance tests.

2.3 Scope

The project scope covers logical network design, FortiGate firewall configuration (HA, LDAP, IPsec VPN, SD-WAN, and firewall policies), and functional testing. The environment is fully virtualized inside GNS3 for lab and academic purposes; no production banking data or real customer systems are involved.

3. Tools and Technologies

Tool / Technology	Purpose in This Project
GNS3	Network emulation platform used to build and interconnect the virtual topology.
FortiGate VM64-KVM (FortiOS)	Next-generation firewall appliances providing routing, HA, VPN, SD-WAN, and LDAP-based authentication at HQ and both branches.
VPCS (Virtual PC Simulator)	Lightweight virtual end-hosts used to represent employee PCs at each site for connectivity testing.
SecureCRT	Terminal/console client used to access and test VPCS and firewall CLI sessions.
TightVNC Viewer	Remote desktop client used to access the FortiGate web-based GUI (FortiOS Administration Console).
Windows Server (DC-1) + Active Directory	Domain Controller providing the LDAP directory service used for centralized authentication.
ISP Cloud / Simulated Internet	Represents the public transport network (192.168.122.0/24) interconnecting HQ, Libya, and Tunisia.

4. Network Topology and Design

The network consists of three sites interconnected through a simulated ISP cloud. The Headquarters (Egypt) site hosts the HA firewall cluster, the internal Domain Controller, and local employee PCs. Two branch sites — Libya and Tunisia — each connect to Headquarters through a dedicated FortiGate branch firewall (BR-1 and BR-2 respectively) over the ISP.

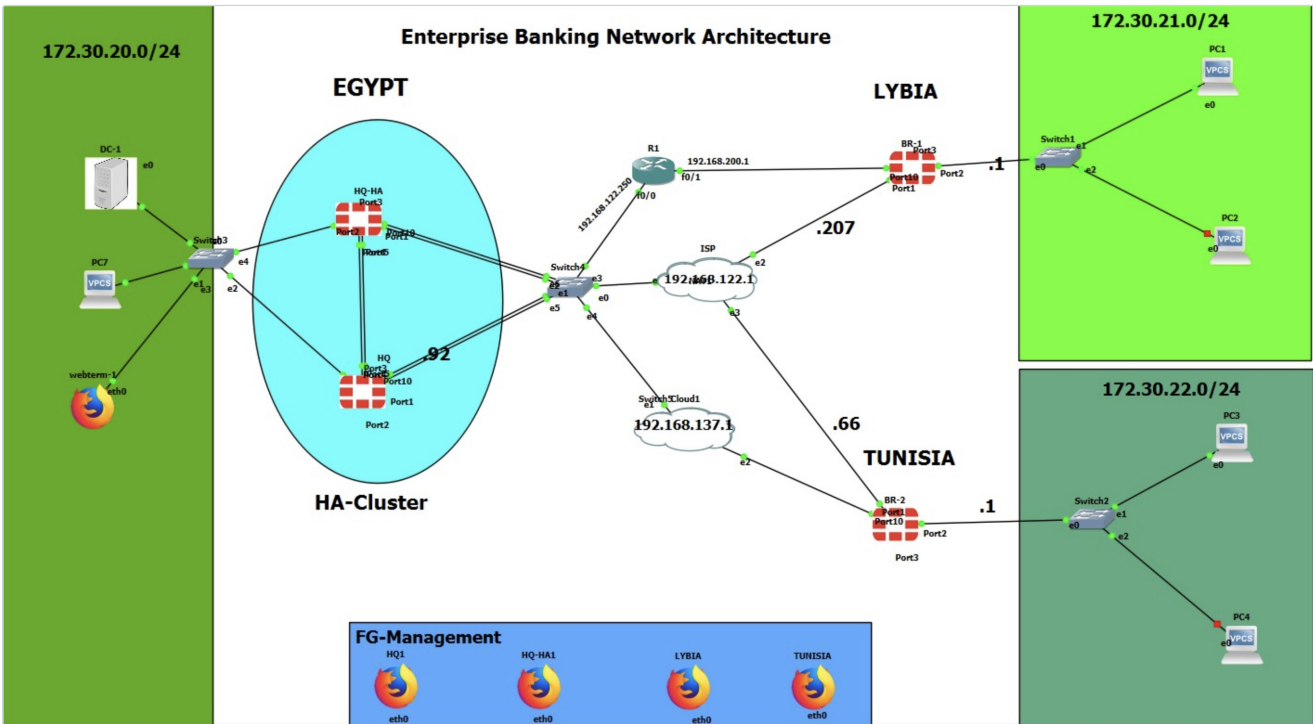


Figure 4.1 — Enterprise Banking Network Architecture: Headquarters (Egypt) HA cluster, ISP transit cloud, and Libya / Tunisia branch sites.

4.1 IP Addressing Plan

Site / Segment	Subnet	Key Devices
Headquarters LAN (Egypt)	172.30.20.0/24	DC-1 (Domain Controller / LDAP), PC7, webterm-1
Libya Branch LAN	172.30.21.0/24	PC1, PC2, Switch1, BR-1
Tunisia Branch LAN	172.30.22.0/24	PC3, PC4, Switch2, BR-2
HQ WAN / Internet Edge	192.168.122.0/24	R1 (edge router), ISP cloud
Libya WAN Link	192.168.122.207	BR-1 (Port2) toward ISP
Tunisia WAN Link	192.168.122.66	BR-2 toward ISP / Cloud1 (192.168.137.1)
HQ Management	192.168.10.88 / .89	FortiGate GUI management (HQ, HQ-HA, BR-1, BR-2)

4.2 High-Level Design Rationale

- The HQ site uses **two FortiGate firewalls** (HQ and HQ-HA) in a cluster inside an HA-Cluster logical boundary, both connected to redundant switches (Switch3, Switch4) — removing HQ firewall as a single point of failure.
- Branch firewalls (BR-1 in Libya, BR-2 in Tunisia) each terminate an IPsec VPN tunnel back to HQ, forming a Hub-and-Spoke topology with HQ as the hub.

- An FG-Management network provides out-of-band browser access (via Firefox) to the administration console of every firewall: HQ, HQ-HA, LYBIA (BR-1), and TUNISIA (BR-2).

5. High Availability (HA) Implementation

High Availability was configured at the Headquarters site to guarantee continuity of firewall, routing, and VPN services if the primary appliance fails. Two identical FortiGate VM64-KVM appliances were clustered using FortiGate's native **FGCP (FortiGate Clustering Protocol)** in **Active-Passive** mode.

5.1 HA Design

Parameter	Primary (HQ)	Secondary (HQ-HA)
HA Mode	Active-Passive	
Cluster / Group Name	HQ-Cluster	
Device Priority	200 (higher → Master)	128 (Slave)
Monitored Interfaces	HQ-WAN (port1), HQ-LAN (port2)	
Heartbeat Interfaces	port4, port5	
Resulting Role	Master	Slave

The device with the higher priority value (200) is elected **Master** and actively processes all production traffic, while the second unit (priority 128) remains in **Slave** (hot-standby) state, continuously synchronizing configuration and session state over the dedicated heartbeat interfaces. If the monitored WAN or LAN interfaces on the Master fail, or the unit itself goes down, the Slave detects the loss of heartbeat and takes over automatically.

5.2 HA Configuration — Master Unit

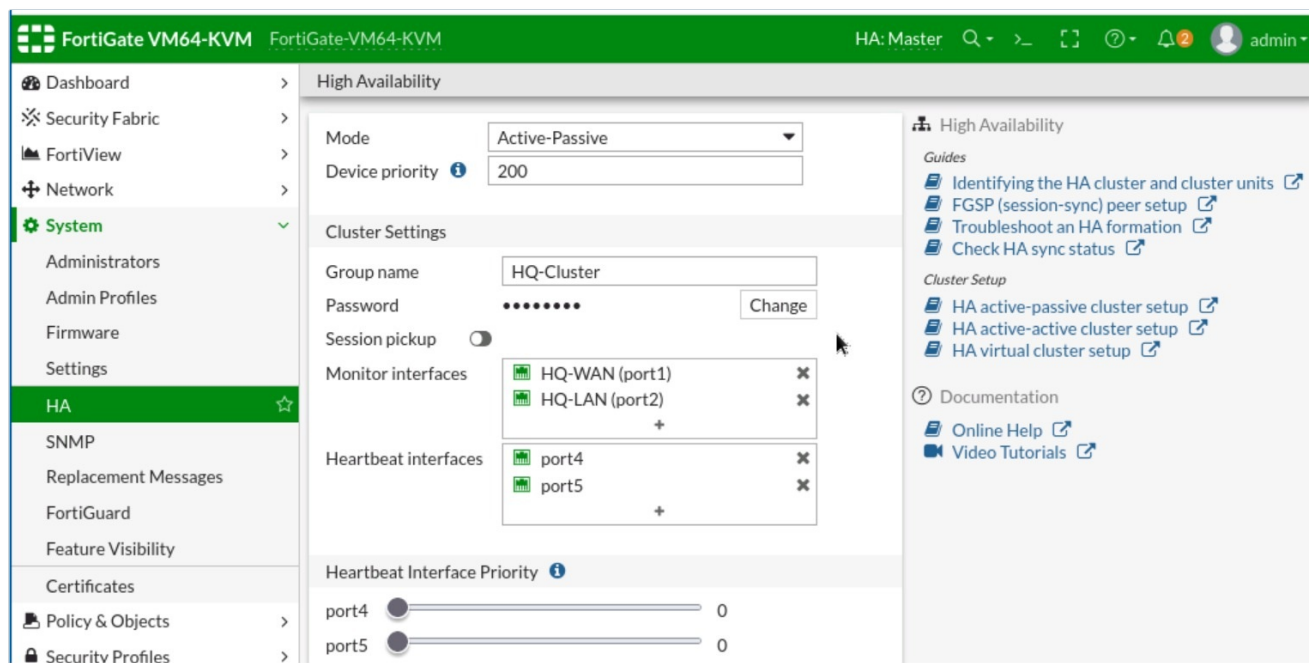


Figure 5.1 — HA configuration page on the Master unit: Active-Passive mode, device priority 200, cluster name "HQ-Cluster", monitored interfaces (port1/port2), and heartbeat interfaces (port4/port5).

5.3 HA Configuration — Secondary Unit

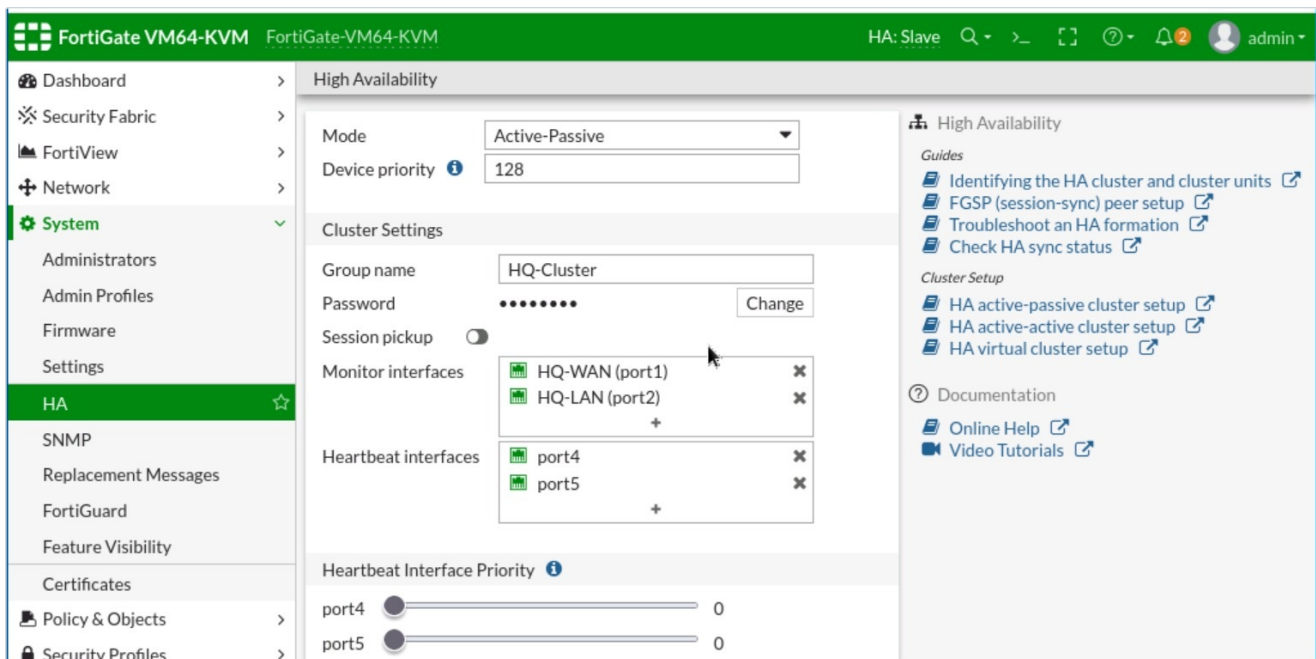


Figure 5.2 — Matching HA configuration on the secondary unit, with device priority 128, joining the same "HQ-Cluster" group so the two units can form a cluster.

5.4 HA Cluster Status Verification

After both units were configured with matching cluster settings, the HA Monitor page confirms that the cluster formed successfully: both members are synchronized, and roles were assigned according to configured priority.

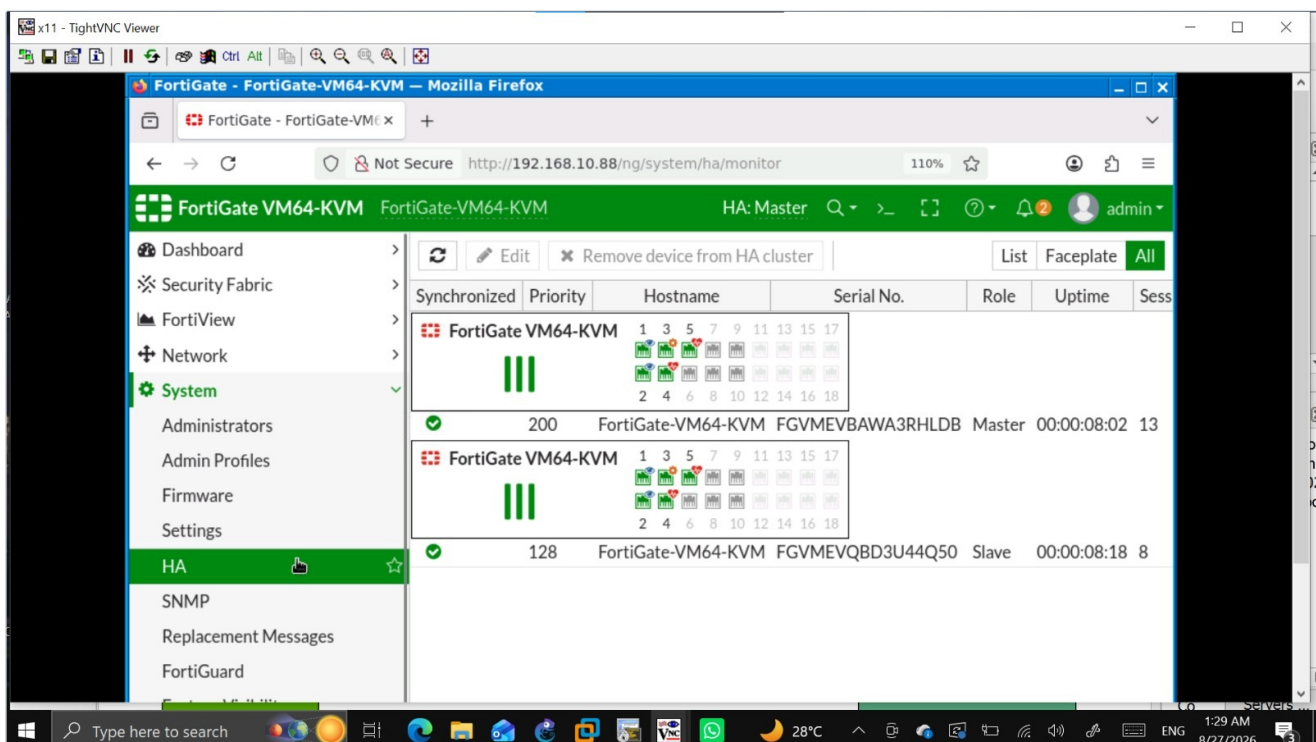


Figure 5.3 — HA Monitor showing two synchronized cluster members: the unit with priority 200 holds the **Master** role, and the unit with priority 128 holds the **Slave** role.

Result: The HA cluster is fully synchronized (green check) with both members reporting healthy status, confirming the Headquarters firewall layer has no single point of failure.

6. Centralized Authentication using LDAP

Rather than maintaining separate local user accounts on the firewall, user authentication is delegated to the organization's existing **Active Directory** domain controller (DC-1, 172.30.20.111) via the **LDAP** protocol. This gives IT administrators a single place to create, disable, or remove employee accounts, with those changes automatically reflected in firewall access control and captive-portal logins.

6.1 LDAP Server Configuration

An LDAP server object was created on the FortiGate pointing to the internal Domain Controller, using standard LDAP port 389 and the `sAMAccountName` attribute as the common name identifier — the standard Windows Active Directory username attribute.

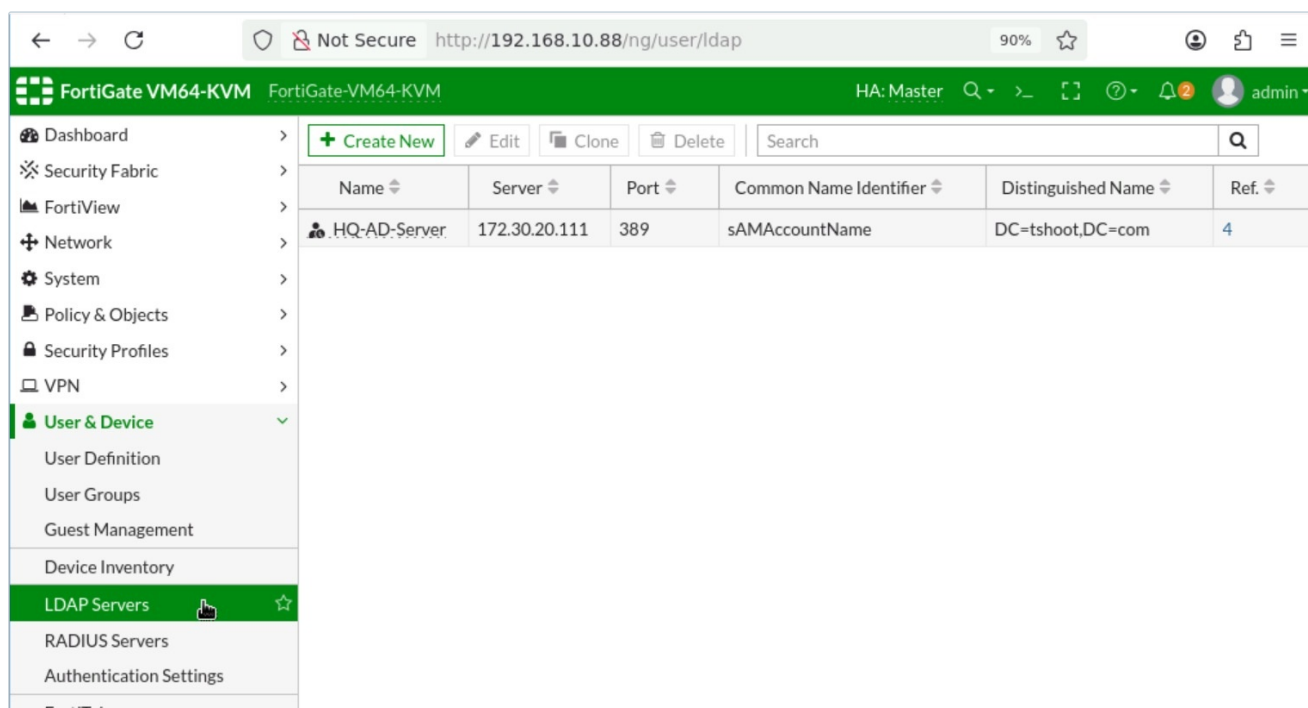


Figure 6.1 — LDAP server object "HQ-AD-Server" configured on the FortiGate, pointing to 172.30.20.111:389, with Distinguished Name `DC=tshoot,DC=com` and common name identifier `sAMAccountName`.

6.2 LDAP-Based User Groups

Firewall user groups were created and linked to the LDAP server so that access policies can be applied by Active Directory group membership rather than by individual user. This includes both LDAP-backed groups and a Fortinet Single Sign-On (FSSO) group for transparent authentication.

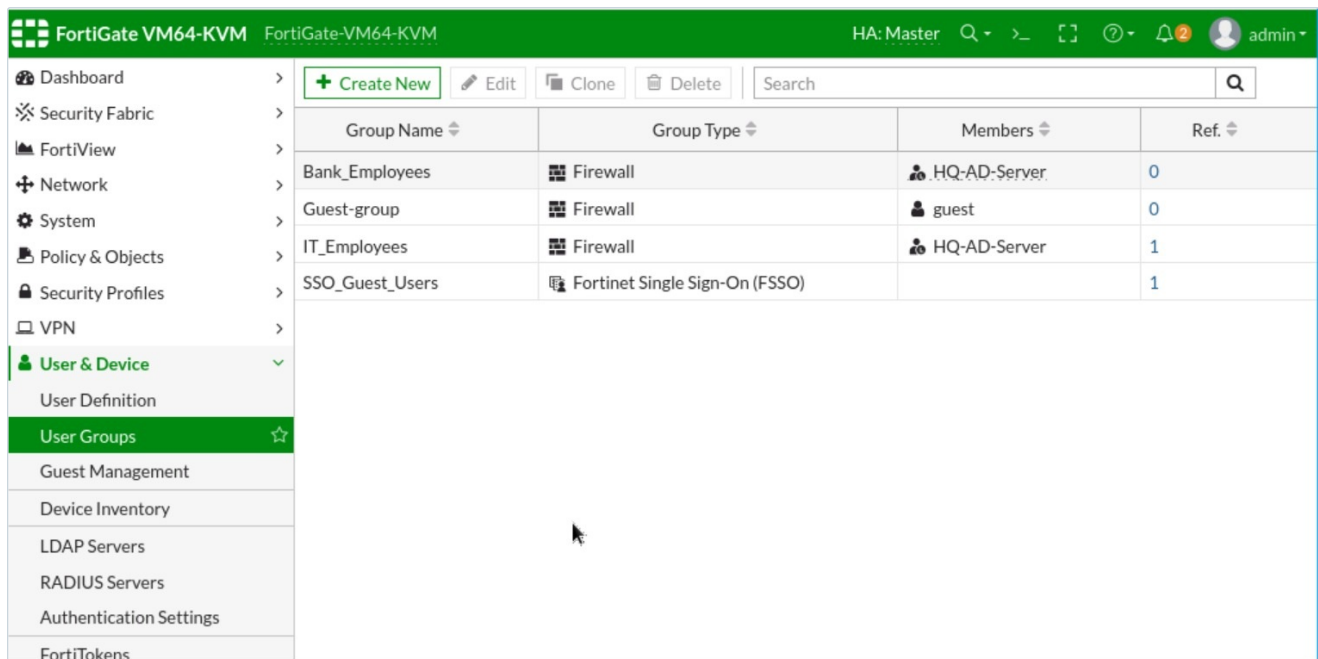


Figure 6.2 — FortiGate User Groups: "Bank_Employees" and "IT_Employees" are Firewall-type groups backed by the HQ-AD-Server LDAP directory; "SSO_Guest_Users" is configured for Fortinet Single Sign-On.

6.3 Enforcing LDAP Authentication in Firewall Policy

The `IT_Employees` LDAP group was attached directly as a source object in the LAN-to-WAN firewall policy (see Figure 7.3, Section 7), meaning only users who successfully authenticate against Active Directory as members of that group are permitted outbound Internet access — enforcing identity-based access control at the network edge.

6.4 Authentication Test

To validate the end-to-end authentication flow, a client on the HQ LAN attempting to reach the Internet is redirected to a FortiGate captive portal login page and prompted for Active Directory credentials before access is granted.

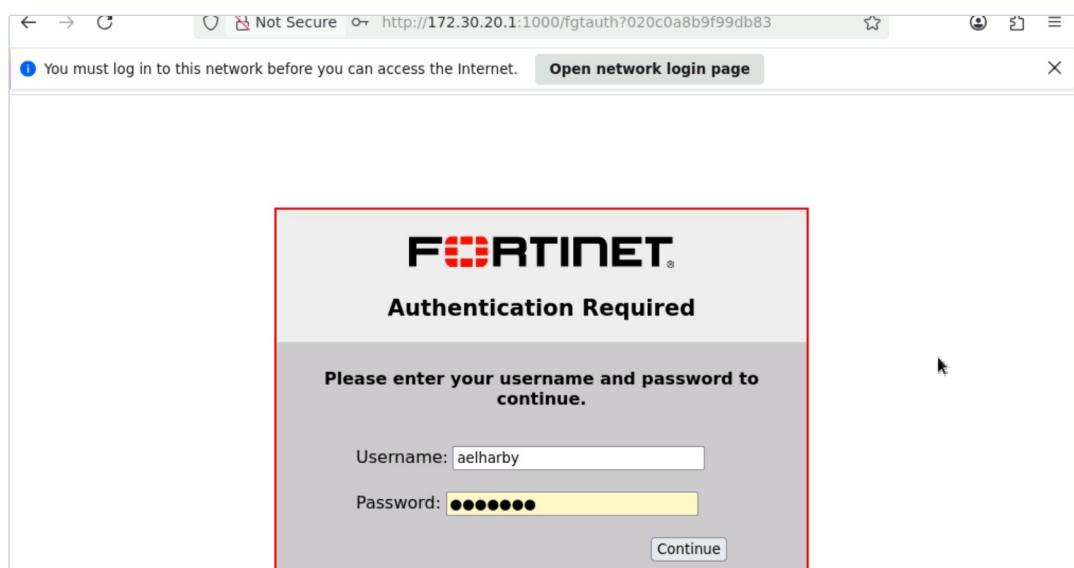


Figure 6.3 — FortiGate captive portal authentication page. The client is redirected to `http://172.30.20.1:1000/fgtauth` and must supply valid Active Directory (LDAP) credentials, e.g. user "aelharby", before Internet access is granted.

Result: The FortiGate successfully queries the LDAP directory in real time, confirming that centralized, directory-based authentication is fully functional and enforced by firewall policy.

7. Site-to-Site VPN (Hub-and-Spoke IPsec)

All traffic between Headquarters and the two branch sites travels across the public ISP cloud (192.168.122.0/24), so it must be encrypted in transit. A **Hub-and-Spoke IPsec VPN** was configured with the HQ firewall cluster acting as the Hub, and the Libya (BR-1) and Tunisia (BR-2) branch firewalls acting as Spokes.

7.1 VPN Topology

Tunnel	Hub (HQ)	Spoke (Remote Gateway)
HQ_VPN_0	HQ Cluster	BR-1 — Libya (192.168.122.207)
HQ_VPN_1	HQ Cluster	BR-2 — Tunisia (192.168.122.66)

7.2 IPsec Tunnel Status

The IPsec Monitor on the HQ firewall confirms both spoke tunnels are established and actively passing traffic, under the "Hub-and-Spoke - FortiGate (Hub)" phase-1 group.

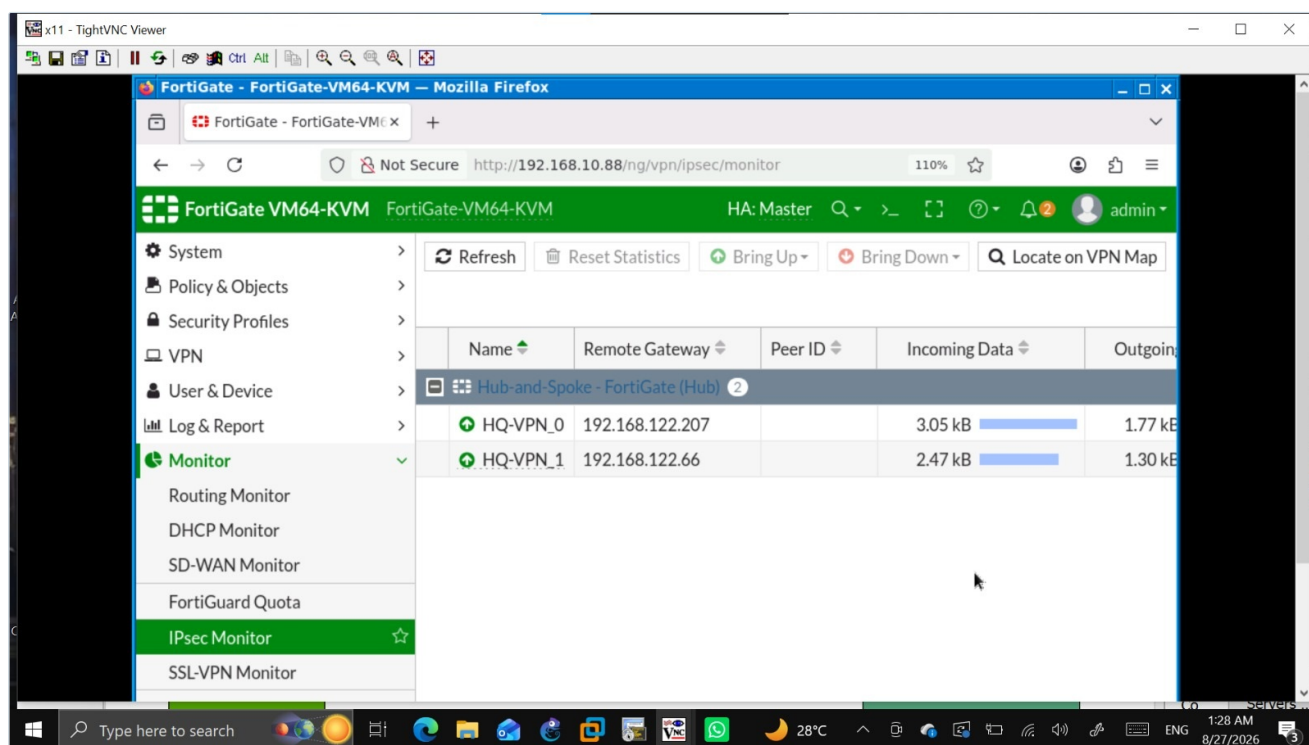


Figure 7.1 — IPsec Monitor showing both spoke tunnels up: HQ-VPN_0 to Libya (192.168.122.207) and HQ-VPN_1 to Tunisia (192.168.122.66), each actively exchanging incoming/outgoing traffic.

7.3 VPN Firewall Policies

Dedicated firewall policies control traffic direction across the VPN: LAN-to-tunnel, tunnel-to-LAN, and spoke-to-spoke communication (allowing Libya and Tunisia branches to reach each other through the HQ hub). The outbound LAN-to-WAN policy is also where the LDAP-authenticated `IT_Employees` group is enforced.

ID	Name	Source	Destination	Schedule	Service	Action	NA
HQ-LAN (port2) → HQ-WAN (port1) 1							
5	LAN-WAN	all IT_Employees	all	always	ALL	✓ ACCEPT	✓ En
HQ-LAN (port2) → MY-Tunnel 1							
3	vpn_MY-Tunnel_local	MY-Tunnel_local	MY-Tunnel_remote	always	ALL	✓ ACCEPT	✗ Dis
HQ-VPN → HQ-LAN (port2) 1							
1	vpn_HQ-VPN_spoke2hub	all	HQ-VPN_local	always	ALL	✓ ACCEPT	✗ Dis
HQ-VPN → HQ-VPN 1							
2	vpn_HQ-VPN_spoke2spoke	all	all	always	ALL	✓ ACCEPT	✗ Dis
MY-Tunnel → HQ-LAN (port2) 1							
4	vpn_MY-Tunnel_remote	all	all	always	ALL	✓ ACCEPT	✗ Dis
Implicit 1							

Figure 7.2 — FortiGate IPv4 Policy table: LAN→WAN policy scoped to the IT_Employees LDAP group, plus VPN policies for HQ-LAN↔MY-Tunnel and spoke-to-spoke (HQ-VPN→HQ-VPN) traffic, all set to ACCEPT.

7.4 VPN Connectivity Testing

Connectivity across the encrypted tunnels was verified from client machines at both branch sites.

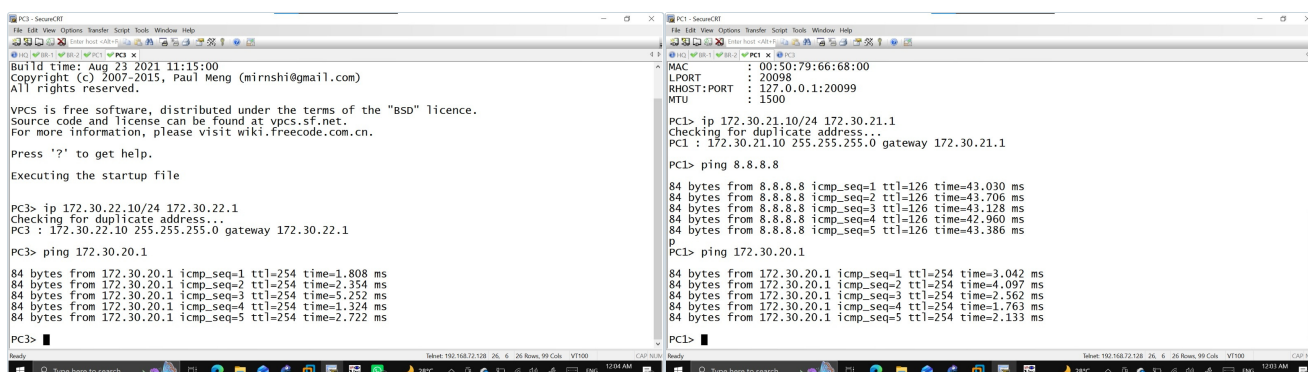


Figure 7.3 — PC3 (Tunisia, 172.30.22.10) successfully pings the HQ gateway (172.30.20.1) across the VPN tunnel — 0% packet loss.

Figure 7.4 — PC1 (Libya, 172.30.21.10) successfully pings both the public Internet (8.8.8.8) and the HQ gateway (172.30.20.1) across the VPN tunnel.

Result: Both branch sites have full, encrypted reachability to Headquarters, and Internet access is also available through the HQ hub — confirming correct VPN routing and NAT configuration.

8. SD-WAN Implementation

SD-WAN was configured on the branch firewalls to continuously measure WAN link quality and support intelligent traffic steering across available paths (port1 and port10), rather than relying on a single static route to Headquarters.

8.1 Performance SLA Monitoring

A Performance SLA profile named `Check_HQ` was created, using 192.168.122.92 (the HQ-side address) as the detect server. It continuously measures packet loss, latency, and jitter on both member links (port1 and port10) and compares them against a configured failure threshold.

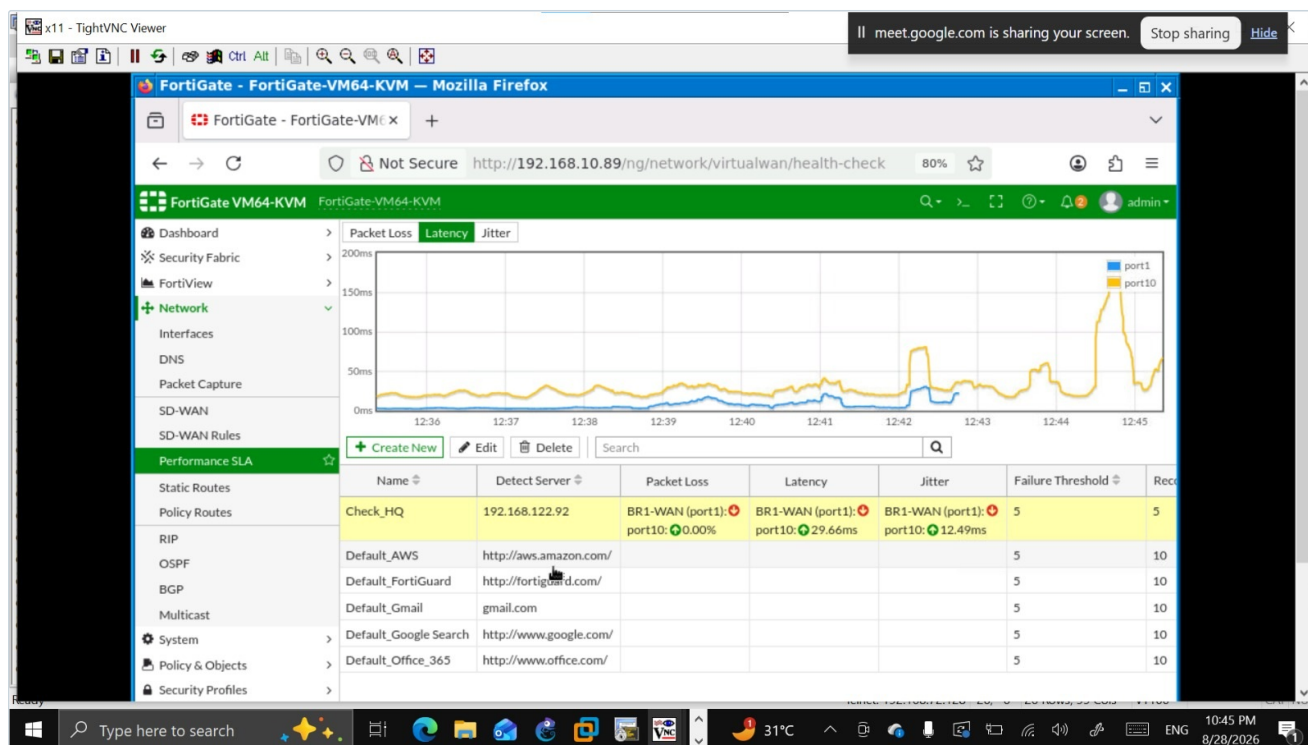


Figure 8.1 — SD-WAN Performance SLA dashboard: the "Check_HQ" health-check probe measures latency on port1 and port10 toward 192.168.122.92, alongside default SLA probes for AWS, FortiGuard, Gmail, Google Search, and Office 365.

8.2 Link Degradation / Failover Behavior

To validate SD-WAN's monitoring behavior under real-world conditions, a continuous ping was run from a branch client to the HQ-side monitored address (192.168.122.92) over an extended period. The results show normal latency most of the time, with visible latency spikes and two consecutive request timeouts — representative of a transient WAN degradation event that the SD-WAN Performance SLA is designed to detect.

```
PC1 - SecureCRT
File Edit View Options Transfer Script Tools Window Help
meet.google.com is sharing your screen. Stop sharing Hide
HQ HQ-HA BR-1 BR-2 R1 PC1 x
84 bytes from 192.168.122.92 icmp_seq=70 ttl=254 time=7.221 ms
84 bytes from 192.168.122.92 icmp_seq=71 ttl=254 time=47.063 ms
84 bytes from 192.168.122.92 icmp_seq=72 ttl=254 time=12.861 ms
84 bytes from 192.168.122.92 icmp_seq=73 ttl=254 time=13.219 ms
84 bytes from 192.168.122.92 icmp_seq=74 ttl=254 time=5.028 ms
84 bytes from 192.168.122.92 icmp_seq=75 ttl=254 time=4.489 ms
84 bytes from 192.168.122.92 icmp_seq=76 ttl=254 time=10.536 ms
84 bytes from 192.168.122.92 icmp_seq=77 ttl=254 time=10.145 ms
84 bytes from 192.168.122.92 icmp_seq=78 ttl=254 time=8.581 ms
84 bytes from 192.168.122.92 icmp_seq=79 ttl=254 time=20.186 ms
84 bytes from 192.168.122.92 icmp_seq=80 ttl=254 time=51.590 ms
84 bytes from 192.168.122.92 icmp_seq=81 ttl=254 time=619.742 ms
84 bytes from 192.168.122.92 icmp_seq=82 ttl=254 time=64.192 ms
192.168.122.92 icmp_seq=83 timeout
192.168.122.92 icmp_seq=84 timeout
84 bytes from 192.168.122.92 icmp_seq=85 ttl=253 time=91.892 ms
84 bytes from 192.168.122.92 icmp_seq=86 ttl=253 time=74.062 ms
84 bytes from 192.168.122.92 icmp_seq=87 ttl=253 time=39.140 ms
84 bytes from 192.168.122.92 icmp_seq=88 ttl=253 time=32.889 ms
84 bytes from 192.168.122.92 icmp_seq=89 ttl=253 time=16.981 ms
84 bytes from 192.168.122.92 icmp_seq=90 ttl=253 time=26.095 ms
84 bytes from 192.168.122.92 icmp_seq=91 ttl=253 time=29.432 ms
84 bytes from 192.168.122.92 icmp_seq=92 ttl=253 time=30.201 ms
84 bytes from 192.168.122.92 icmp_seq=93 ttl=253 time=23.256 ms
84 bytes from 192.168.122.92 icmp_seq=94 ttl=253 time=17.153 ms
84 bytes from 192.168.122.92 icmp_seq=95 ttl=253 time=50.158 ms
Ready Telnet: 192.168.72.128 26 6 26 Rows, 99 Cols VT100 CAP NUM
Type here to search 31°C 10:45 PM 8/28/2026
```

Figure 8.2 — Extended ping test to 192.168.122.92: normal round-trip times (5-50 ms) with a brief period of high latency (up to ~620 ms) and two timed-out packets (sequence 83-84), captured by the SD-WAN health-check as a degraded-link event.

Observation: The temporary spike in latency and packet loss shown above is exactly the type of condition the `Check_HQ` Performance SLA (Figure 8.1) is configured to detect, allowing SD-WAN rules to steer new sessions to the healthier link once the failure threshold is crossed.

9. Testing and Results

#	Test	Method	Result
1	HA cluster formation	HA Monitor page on both units	Pass — Master (priority 200) / Slave (priority 128) synchronized
2	LDAP directory binding	LDAP server object test / captive portal login	Pass — user "aelharby" authenticated via AD
3	Group-based access policy	IT_Employees LDAP group in LAN-WAN policy	Pass — only authenticated group members get outbound access
4	Site-to-site VPN (Libya)	Ping HQ from PC1	Pass — 0% loss to 172.30.20.1
5	Site-to-site VPN (Tunisia)	Ping HQ from PC3	Pass — 0% loss to 172.30.20.1
6	Internet access via HQ hub	Ping 8.8.8.8 from PC1	Pass
7	IPsec tunnel status	IPsec Monitor	Pass — both spoke tunnels up and passing traffic
8	SD-WAN SLA monitoring	Performance SLA dashboard	Pass — latency/jitter/packet loss tracked per link
9	WAN degradation detection	Extended ping to 192.168.122.92	Pass — latency spike and timeouts correctly captured

10. Challenges and Solutions

- **HA synchronization mismatches:** Initial cluster formation requires both FortiGate units to run identical firmware and matching HA password/group name; any mismatch prevents the cluster from forming. This was resolved by verifying firmware versions and cluster settings on both units before enabling HA.
- **LDAP bind failures:** Early authentication attempts failed until the correct base Distinguished Name (`DC=tshoot,DC=com`) and common name identifier (`sAMAccountName`) were set to match the Active Directory schema.
- **Asymmetric routing across VPN spokes:** Spoke-to-spoke traffic between Libya and Tunisia required an explicit HQ-VPN → HQ-VPN policy in addition to the standard hub policies, since traffic between two spokes is not implicitly allowed.
- **Transient WAN latency:** The extended ping test surfaced brief high-latency periods and dropped packets on the simulated ISP link, reinforcing the need for the SD-WAN Performance SLA to detect and react to such conditions automatically.

11. Conclusion and Future Work

This project successfully demonstrates a practical, defense-in-depth network security architecture suitable for a distributed banking environment. By combining **FortiGate High Availability** for resilience, **LDAP-integrated authentication** for centralized identity control, an encrypted **Hub-and-Spoke IPsec VPN** for confidential inter-site communication, and **SD-WAN performance monitoring** for WAN resilience, the resulting network satisfies the core availability, confidentiality, and manageability requirements expected of a financial institution's infrastructure.

Future enhancements to this project could include: migrating the HA cluster to Active-Active mode for load distribution, adding multi-factor authentication (MFA) on top of LDAP for privileged accounts, extending SD-WAN rules to perform automatic application-aware traffic steering, and integrating centralized logging (FortiAnalyzer or a SIEM) for security monitoring across all four firewalls.

12. References

- Fortinet Inc., *FortiOS Administration Guide* — High Availability, LDAP, IPsec VPN, and SD-WAN chapters.
- Fortinet Documentation Library, *FortiGate HA (FGCP) Cluster Setup Guide*.
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- GNS3 Documentation, *GNS3 User Guide* — network emulation platform used for the lab topology.
- Microsoft Docs, *Active Directory Lightweight Directory Services (LDAP) Overview*.

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